

**LAB ORIENTED PROJECT
REPORT**
ON
EDUCATIONAL KIT PREPERATION

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1 Abstract

In this era of globalization use of a teaching kit plays a vital role for the performance of students in primary education[1]. Research indicates that when learners actively participate in the learning process through hands-on experiences, their understanding becomes more meaningful and impactful. Students remember only a fraction of what they hear but majority of what they do[2]. This form of experiential learning not only deepens comprehension but also promotes critical thinking and problem-solving skills, essential for mastering complex scientific and mathematical concepts. There is a statistically significant improvement and difference in both conceptual understanding and problem solving skills among the students who are taught with the help of a realist hands-on learning approach. It provides students with an opportunity to learn practically[3].

2 Introduction

The motivation for taking up this project stems from the need to make abstract concepts in subjects like geometry, physics, and mathematics more accessible and engaging for students. Traditional classroom instruction often struggles to convey these ideas in a way that resonates with all learners. By developing innovative, interactive tools such as puzzle-based models, we aim to create educational kits that offer students a tangible way to explore and understand theoretical concepts. This hands-on approach not only enhances understanding but also fosters a sense of curiosity and inquiry, which is essential for lifelong learning.[4] By incorporating real-life demonstrations into classroom learning, we aim to improve learning outcomes by enhancing student engagement, motivation, and retention of knowledge.[5]

Our project focuses on the conceptualization, design, and production of an innovative educational science toolkit that leverages interactive, hands-on methods to enhance comprehension and engagement in science, mathematics, and engineering. Recognizing the challenges posed by traditional teaching techniques in conveying abstract concepts, this project employs cutting-edge technologies, including laser cutting and 3D printing, to develop precise, cost-effective, and scalable educational tools. These tools are designed to bridge the gap between theoretical knowledge and practical understanding, fostering critical thinking and active learning in alignment with modern STEM education goals. Freeman et al. (2014) conducted a meta-analysis across 225 studies and found that active learning reduces failure rates in STEM disciplines by 55% and increases exam performance by 6% [6].

The success of this project underscores the importance of experiential learning in STEM education. By integrating tangible tools into curricula, educators can make abstract concepts more accessible, enhance learning outcomes, and inspire a new generation of critical thinkers and innovators. This toolkit serves as a model for incorporating modern technologies and active learning strategies into classroom settings, contributing to the evolution of teaching practices in the 21st Century.

3 Equipment

BITS Pilani, KK Birla Goa Campus has many facilities which give students the opportunities to work with many different kinds of equipment. The Sandbox Lab has 3D printer which can be used to produce solid products, it also has many electronic equipment which can be used to make devices and projects which can help students in their personal projects as well as projects of clubs and departments. The Innovation Lab has the Laser cutting machine, which can be used to produce 2D designs and give shape to innovative ideas.

The Mechanical Workshop has many equipment which includes both traditional and non traditional manufacturing, they can be used to make projects with vast range of applications. For instance lathes, milling machines, and CNC equipment to create precise parts for various industries. Producing custom parts or prototypes for engineering projects using CNC machines or manual machining. Creating custom jigs and fixtures that can help in woodworking or other specialized projects. With the right skills and marketing, a mechanical workshop can serve a wide range of customers, from individual vehicle owners to industrial clients.

For our the first few ideas of our project, we used the laser cutting machine to make different puzzles. A laser cutting machine is a precision tool that uses a focused, high-powered laser beam to cut, engrave, or etch materials like wood, metal, acrylic, plastic, and fabric. The laser operates by directing a concentrated beam of light through an optical system onto the surface of the material, where it heats, melts, burns, or vaporizes the material along a pre-programmed path. The machine is typically controlled by a Computer Numerical Control (CNC) system, which follows a digital design file (such as a .dxf) to create intricate cuts or engravings with high accuracy.

The key components of a laser cutting machine are:-

- 1.Laser Source: Generates the laser beam. It includes CO2 lasers for non-metal materials and fiber lasers for metals.
- 2.Optics System: Mirrors and lenses focus and direct the laser beam onto the cutting surface.
- 3.CNC Software: Software that directs the laser's movement according to the design file, ensuring precise cuts and engravings. We used LaserCAD for our project because it was the software compatible with the machine we were using.
- 4.Material Bed: The material to be cut is placed here.

It's working is based on thermal separation process. The laser beam heats the material at the focal point, causing it to melt, burn, or vaporize, depending on the material's properties and thickness. The result is a clean, narrow cut with minimal waste and highly detailed edges.

The other projects were made using the 3D printer available in the Sandbox Lab at our institute. A 3D printing machine is an advanced tool in additive manufacturing that constructs three-dimensional objects layer by layer based on a digital model. This process begins with creating a design using CAD software or scanning an existing object to generate a blueprint. The model is then processed through slicing software, which divides it into thin horizontal layers and generates G-code instructions to guide the printer. During printing, the machine deposits material—such as thermoplastic filaments, liquid resin, or metal powders—layer by layer to form the object. The type of material and method used depend on the technology, which can include Fused Deposition Modeling (FDM) for melting and extruding plastics, Stereolithography (SLA) for curing resin with UV light, or Selective Laser Sintering (SLS) for fusing powder materials using a laser. Advanced techniques like Direct Metal Laser Sintering (DMLS) and Binder Jetting enable the creation of durable metal parts and detailed ceramic models, expanding the range of applications. The workflow involves digital design creation, file preparation (.stl for 3D Printing), and machine operation monitoring.

3D printers are highly versatile and find extensive use in industries such as healthcare, aerospace, automotive, and education. They enable rapid prototyping for product development, custom manufacturing for personalized medical devices like prosthetics and dental implants, and small-scale production of intricate designs. Its benefits include the ability to produce complex geometries, reduce material waste, and facilitate on-demand manufacturing without the need for extensive inventory.

4 Procedure

4.1 Ideation

For the idea for this project, we went through some videos and websites sent by our professor, and also we searched on the internet for more references. After that we come up with these ideas:

1. Triangle Inequality Puzzle: This is based on the property of a triangle that “The sum of any two sides of a triangle is greater than the third side”.

2. Area of Trapezium: This is depicting the formulation of Area of Trapezium which is:

$$\text{Area} = \frac{1}{2} (a + b) h$$

where, a and b = bases of the trapezium (parallel sides), and, h = height (the perpendicular distance between a and b)

3. Tangram Puzzle: The Tangram is made by cutting a square into seven pieces. The puzzle lies in using all seven pieces of the Tangram to make birds, houses, boats, people and geometric shapes. In each case you have to use all the seven pieces - no more, no less.

4. 4-Bar Linkage: A four-bar linkage is a mechanical linkage that consists of four rigid bars connected by joints to form a closed loop. It’s the simplest movable closed-chain linkage and is widely used in many applications.

5. Engine Piston: This small setup explains the working of a 4-stroke single cylinder engine piston. It is made up of the piston head, piston arm, crank, crankshaft and crankcase. This model is unique in the sense that it can be setup without manufacturing complicated parts and provides a simplified cross-sectional view of the engine as opposed to the actual model.

6. Dissection Puzzle: This puzzle shows how a square can be converted into an equilateral triangle by cutting various shapes and rejoining them. This proves that the area of both shapes is the same but the perimeter is different.

4.2 Design and Manufacturing

4.2.1 Laser Cutting

Laser cutting machines can work on materials like wood (for natural, tactile products), acrylic (for sleek, colorful designs), metal (for durability), and fabric or leather. For our project we selected acrylic sheets as our work material. Since the laser operates in two dimensions (for cutting or engraving), the design must be created as a 2-D vector file. We prepared our designs according to suitable dimensions using AutoCAD software. These designs were downloaded

as (.dxf) files and imported onto another software called LaserCAD. This software translates the design into instructions for the laser cutter. We adjusted the laser settings (power, speed, focus) based on the material's thickness and type. Thicker materials require more power or slower cutting speeds. After inputting the parameters and verifying dimensions, we downloaded the file in pwj5 format and loaded them in a pendrive. This pendrive was then attached to the laser cutting machine.

We placed the acrylic sheet onto the laser bed and set the zero(origin) for the laser nozzle. We tested the power setting by cutting out a simple square design. Then we loaded our design files for each puzzle one by one, changing the origin each time to ensure enough material is available for cutting all dimensions of the puzzle correctly. Once the 3 puzzles were cut out, we removed them from the machine and made them ready for presentation.

4.2.2 3D-Printing

We prepared the 3D CAD models which had to be printed. Dimensioning and tolerances were taken care of so as to have a perfectly sized and working model. The CAD model was exported in STL format and sliced using Cura software. This generated the G-Code for the 3D printer on which the nozzle would set coordinates and start printing. The material spool keeps providing PLA wire which is drawn by the machine and deposited by melting.

The machine is operated on 0.1 mm least count. Once the model is ready, it can be removed from the machine and cleaned and examined for defects. The clearances are verified and parts of the project are assembled.

5 Results

After manufacturing we displayed the projects and verified their geometric properties and functions. Models of the projects are depicted below.

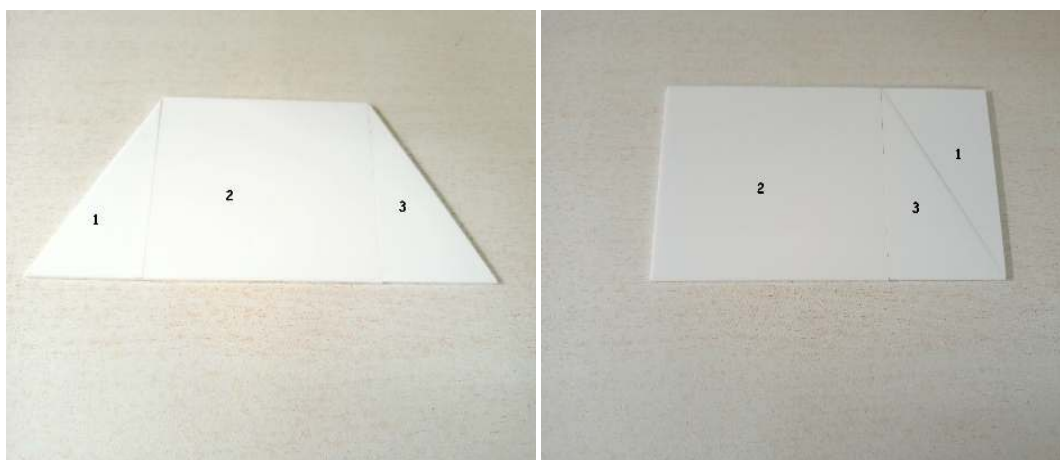


Triangle and its sides

Comparison of lengths of sides

Figure 1: Triangle Inequality

The Triangle Inequality Puzzle is a hands-on tool that teaches the Triangle Inequality Theorem, which states that the sum of any two sides of a triangle must be greater than the third side. It enhances geometric understanding by introducing fundamental concepts of triangle formation as students explore the limits of shape construction. It is an amazing puzzle which introduces them to the basic concepts of perimeter and area and how it can be manipulated accordingly. By testing different combinations of sides, students develop logical thinking and critical analysis, which are applicable in real-world scenarios like navigation and network design.

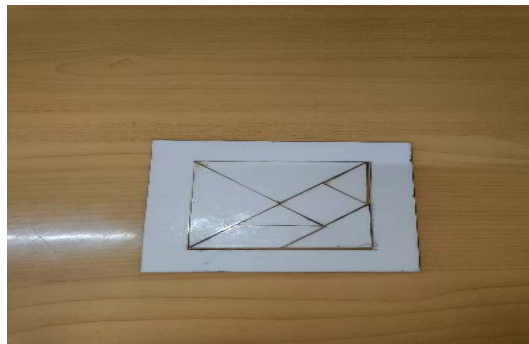


Trapezium

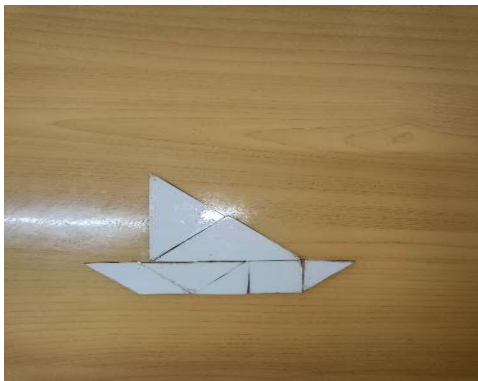
Rearranged as a rectangle

Figure 2: Area of Trapezium

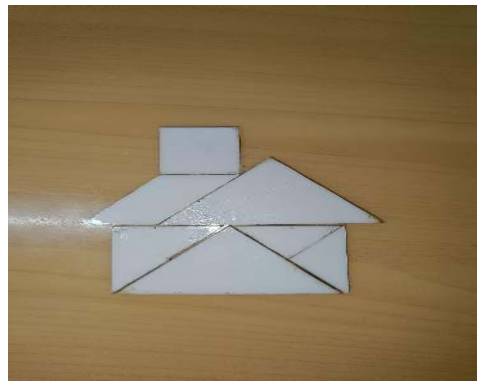
This helps students explore geometric concepts like shape properties, congruence, perimeter, and area. It enhances concepts of geometric transformations like rotation and reflection. The activity also encourages discussions on symmetry and improves understanding of geometric relationships. It helps us understand how geometrical shapes of different area and perimeter can be combined together to form another geometrical shape of importance and vice-versa. These skills are crucial in fields like engineering and architecture.



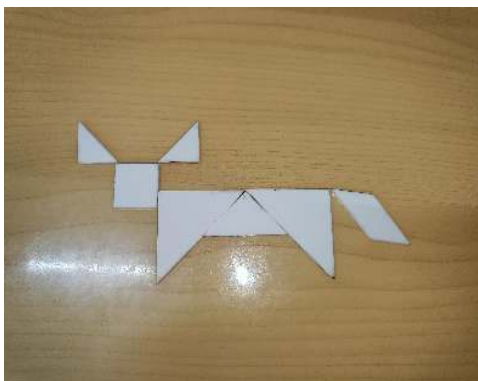
Tangram



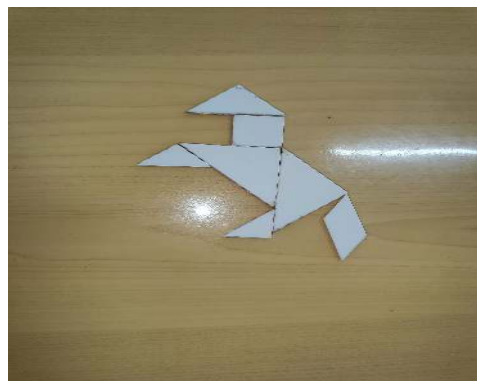
Boat



House



Fox



Horse

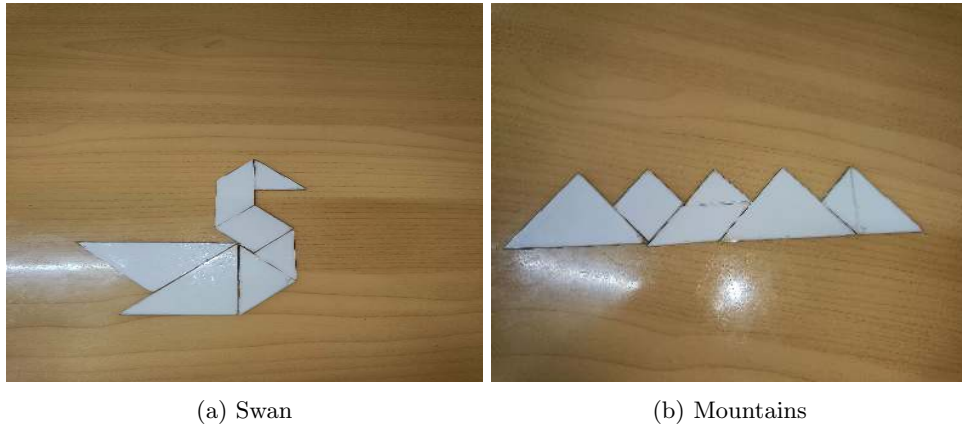


Figure 3: Tangram Puzzle

Tangram puzzles are a tool for enhancing cognitive, spatial, and problem-solving skills. They help develop spatial awareness, geometric reasoning, and math skills (fractions, shapes, and areas). Working with the pieces introduces children to concepts like symmetry, congruency, and geometry, aiding in a deeper understanding of mathematical principles. Tangrams foster creativity, encouraging artistic expression by forming different shapes and patterns. They improve logical thinking, hand-eye coordination, and fine motor skills as students manipulate the pieces to form various shapes which we observe in our surroundings. They promote patience, perseverance, and teamwork in group settings. The potential outcomes include improved critical thinking, visualization skills, and a stronger foundation in STEM learning. It also helps children in observing various geometrical shapes in our surroundings leading to a better understanding of figures and shapes.

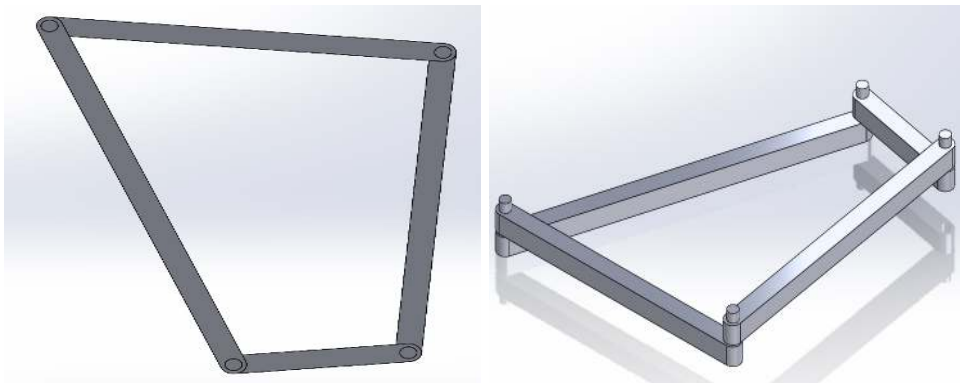


Figure 4: 4-Bar Linkage Mechanism

This 3D-printed 4-bar linkage mechanism demonstrates the fundamental principles of planar motion. The mechanism consists of four rigid links connected by pivot joints to form a closed chain. Depending on the configuration, it can function as a crank-rocker, double-crank, or double-rocker system, among others. Such linkages are widely used in machines for motion transmission and mechanical advantage, forming the basis of many engineering and

industrial applications.



Figure 5: Engine Piston Mechanism

This 3D-printed engine model demonstrates the fundamental components of an internal combustion engine: the piston head, piston arm, crank, crankshaft, and crankcase. These parts work together to convert the linear motion of the piston into the rotational motion of the crankshaft, showcasing the basic mechanics of how engines generate power. The 4 strokes of an engine- intake, compression, combustion and exhaust can be easily demonstrated using this model. It is a compact, feasible and easy to understand.

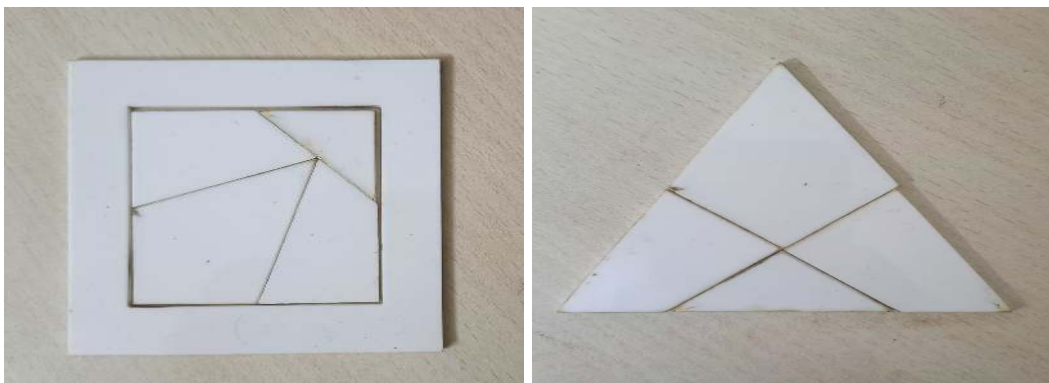


Figure 6: Dissection Puzzle

Here we are converting a square to an equilateral triangle using the same pieces so they have the same area but we observe that the perimeter is different.

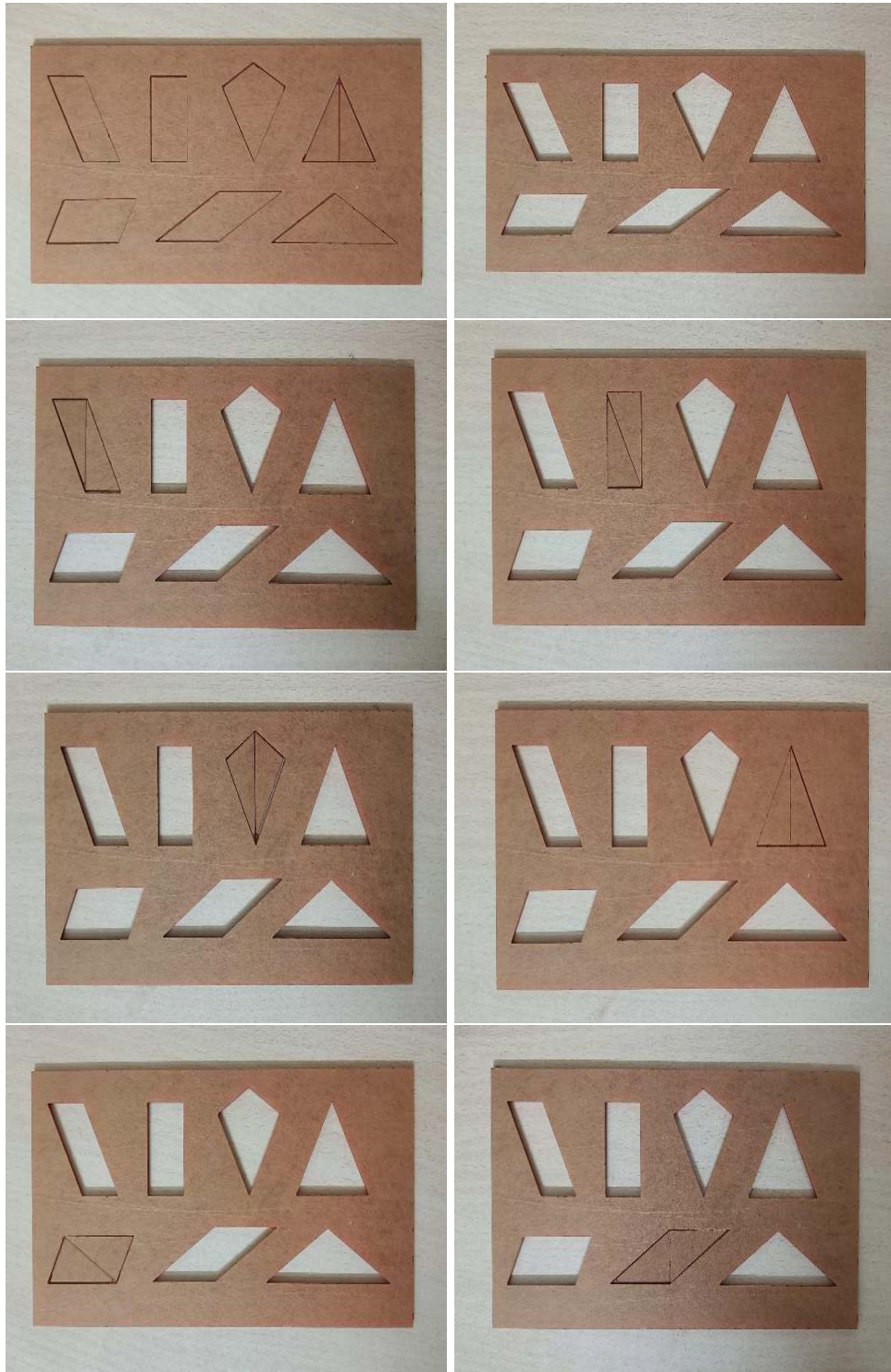


Figure 7: Geometry with two triangles

Here we can learn the concept of perimeter and area of a geometrical shape. We can observe that each geometrical shape here has a different perimeter but all the shapes can be occupied by two congruent triangles which means they have an equal area.

6 Conclusion

All the models designed by us are aimed at designing an educational kit that bridges theoretical concepts with practical applications, making learning more engaging. By enabling students to actively interact with educational materials, the toolkit overcomes the limitations of passive learning, fostering curiosity, creativity, and deeper understanding. Furthermore, the affordability and scalability of the toolkit make it a viable solution for widespread adoption in schools and colleges.

The geometric puzzles including the triangle inequality, tangrams, and the square to equilateral triangle transformation, were designed to develop spatial reasoning and problem-solving skills. Research by Clements and Sarama (2011) found that early spatial reasoning interventions improve problem-solving and geometry test scores by up to 30% [7]. Tangrams are specifically mentioned in studies like those by Mix and Cheng (2012) to improve spatial visualization and mental rotation, abilities critical for fields like engineering and design. They reported that integrating tangram puzzles into classrooms improved spatial visualization skills by 35 % on average.[8] Also, the puzzle demonstrating identical areas with different perimeters introduces students to nuanced geometric relationships, a principle supported by research from Kamii and Kysh (2006), which highlights the importance of understanding mathematical concepts over rote memorization.[9]

The mechanical models, such as the single-cylinder four-stroke engine and 4-bar linkage mechanism, simplify complex engineering principles, making them accessible to students. The 3 of us, along with our peers, personally thought that having these types of models on our campus would help us visualise a lot of concepts in a better way. These models explore the concepts of kinematics and energy transfer, directly addressing the need for hands-on learning in engineering education. Prince (2004) in his article emphasizes that hands-on models significantly improve students' ability to retain and apply engineering concepts. [10]

Technologies like laser cutting and 3D printing enabled the creation of highly accurate, cost-effective models. Blikstein (2013) noted that integrating such technologies into the education system fosters creativity, critical thinking, and design skills, aligning with modern educational goals.[11]

The geometric puzzles and mechanical models not only engage students but also foster critical thinking, spatial reasoning, and practical engineering understanding. All the research papers we read validated the effectiveness of this approach, with improvements in retention, conceptual clarity, and application-oriented learning.

References

- [1] URL: https://pu.edu.pk/images/journal/studies/PDF-FILES/Artical-7_v17-1-16.pdf.
- [2] K. Hawtrey. *Using Experiential Learning Techniques*, The Journal of Economic Education, 2007. URL: <http://dx.doi.org/10.3200/JECE.38.2.143-152>.
- [3] Michael K. Areaya S. Tessema G. *Realist hands-on learning approach and its contributions to learners' conceptual understanding and problem-solving skills on solid geometry*. 2024. URL: <https://doi.org/10.29333/pr/14096>.
- [4] URL: <https://www.reachandteach.net/post/the-power-of-hands-on-experiments-igniting-scientific-curiosity-in-students>.
- [5] URL: <https://flexible.learning.ubc.ca/files/2020/09/experiential-learning-2020.pdf>.
- [6] M. McDonough S. Freeman S.L. Eddy. "Active learning increases student performance in science, engineering, and mathematics". In: *Proc. Natl. Acad. Sci. U.S.A.* (2014). URL: <https://doi.org/10.1073/pnas.1319030111>.
- [7] Sarama J. Clements D.H. "Early Childhood Mathematics Education Research: Learning Trajectories for Young Children". In: (2009). URL: <https://doi.org/10.4324/9780203883785>.
- [8] Cheng YL Mix KS. "The relation between space and math: developmental and educational implications". In: *Adv Child Dev Behav* (2012). URL: <https://doi.org/10.1016/B978-0-12-394388-0.00006-X>.
- [9] Constance Kamii and Judith Kysh. "The difficulty of "length $\times$ width": Is a square the unit of measurement?" In: *The Journal of Mathematical Behavior* 25.2 (2006), pp. 105–115. ISSN: 0732-3123. DOI: <https://doi.org/10.1016/j.jmathb.2006.02.001>. URL: <https://www.sciencedirect.com/science/article/pii/S0732312306000125>.
- [10] URL: https://engr.ncsu.edu/wp-content/uploads/drive/1smSpn4AiHSh8z7a0MHDBwhb_JhcoLQmI/2004-Prince_AL.pdf.
- [11] URL: <https://tltlab.org/wp-content/uploads/2019/02/2013.Book-B.Digital.pdf>.